



LESSON PLAN

Basic Urban Farming with Hydroponics

WSQ Course Code: TGS-2025053916

Version 2.1 · 29 August 2026

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Document Version Control Record

Version	Date	Description	Approved by
1.0	24 March 2025	Legacy approved course package baseline.	Course Development Team
2.0	29 August 2026	Technical rebuild with current Singapore context, aligned Activities and WSQ administration.	Dr Alfred Ang
2.1	29 August 2026	Aligned to the v2.1 deck with authentic hydroponics-practice imagery; delivery sequence and 139-slide ranges retained.	Dr Alfred Ang

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Course Information

Course title	Basic Urban Farming with Hydroponics
Course code	TGS-2025053916
Duration	1 day · 8 hours excluding lunch
TSC	AGR-FOP-2002-1.1 Urban Farming Implementation and Management Control
Assessment	WA-SAQ 30 minutes; PP 30 minutes
Venue	MEOD Farm, 13 Neo Tiew Harvest Lane, Singapore 719020 (verify joining instructions)

Delivery Schedule

Time	Duration	Topic / slides	Facilitation	Learner evidence	Resources
09:00–09:30	0.5 h	Briefing, profiles, access, outcomes and safety · slides 1–17	Questioning; access and equipment-risk examples	Verbal checks	Deck, LG
09:30–10:30	1.0 h	Foundations and systems · slides 18–45	Mechanism diagrams; system comparison	System-selection rationale	Deck, legacy visuals
10:30–10:45	0.25 h	Break	—	—	—
10:45–12:30	1.75 h	Water, nutrients, crop controls and Singapore cases · slides 46–96	Data interpretation; Singapore cases; root diagnosis	pH/EC reasoning; case critique	Deck, meters, LG
12:30–13:30	—	Lunch (not counted)	—	—	—
13:30–14:30	1.0 h	Farm tour · slides 97–106	Guided observation; C/O/I evidence coaching	Completed farm-tour checklist	Activity 01 PDF
14:30–16:00	1.5 h	Hydroponics kit practice · slides 107–118	Demonstrate then coach against SOP; observe A1–A4	Commissioning and correction record	Kit, PPE, meters, Activity 02 PDF
16:00–16:15	0.25 h	Break	—	—	—
16:15–17:00	0.75 h	Aquaponics theory · slides 119–131	Nitrogen-cycle loop; coupled-risk diagnosis	Aquaponics fault explanation	Deck, LG
17:00–18:00	1.0 h	Assessment and close · slides 132–139	Administer WA-SAQ then PP; feedback	Candidate responses and assessor observation	Assessment papers

Final slide ranges are subject to the generated deck inventory (139 slides); the trainer should use the section dividers and slide-map JSON for exact navigation.

Facilitation and Assessment Notes

- Do not present the detailed operating procedure from slides; use the Learner Guide and activity PDFs at the workbench.
- Before the farm tour, confirm site rules, photography permissions, emergency arrangements and learner accessibility needs.
- Use questioning to test why each control exists, not only whether a learner can repeat a setpoint.
- For practical performance, observe safe preparation, operation to target, compliance/housekeeping and cleaning/maintenance.
- Stop any task when equipment, chemical, electrical or hygiene controls are missing.